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SUPERSEDING
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MILITARY SPECIFICATION

ALARM SET, ANTI-INTRUSION, RESTRICTED AREA:

AN/GSQ-159 (DSID)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a hand-emplaced, anti-intrusion seismic detection device.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-C-850	- Cushioning Material, Polystyrene Expanded, Resilient (for Packaging Uses).
PPP-T-97	- Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

MILITARY

MIL-C-45662	- Calibration System Requirements.
MIL-P-60312	- Parts, Molded, Plastic Foam Polystyrene (for Use with Ammunition).

FSC 6350

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-781	- Reliability Tests: Exponential Distribution.
MIL-STD-810	- Environmental Test Methods.

FEDERAL

FED. STD. No. 356	- Commercial Packaging of Supplies and Equipment.

DRAWINGS

ME

TA13218E9400	- Alarm Set, Anti-Intrusion Restricted Area: AN/GSQ-159 (DSID).

(Copies of specifications, standards, and drawings required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The Alarm Set, Anti-Intrusion, Restricted Area: AN/GSQ-159 (hereinafter called DSID) shall be in accordance with Top Assembly TA13218E9400 and as specified herein. The DSID shall operate while implanted in the ground up to its antenna base, so that only the antenna is exposed. The DSID shall be nonrecoverable in that, having been deployed, the DSID cannot be switched off or fitted with a replacement battery and later reused. Provisions are incorporated into

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the DSID to allow a one-time setting of the transmitted identification code, gain, and life timing. The DSID shall consist of two subassemblies, the electronics package, and the power pack.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed materials, dimensions, or tolerances is permissible without prior approval of the contracting officer. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The contractor shall furnish one or more DSID's as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove prior to starting production that his production methods and choice of design detail will produce DSID's that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4; and, unless otherwise specified herein, all examination and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests, as specified (see 6.2).

3.3 Initial production. When specified (see 6.2), the contractor shall furnish to the Government one or more DSID's for inspection as specified in 4.4.

3.4 Material. Material shall be as specified herein and as shown on the applicable drawings. Inherently fungus-resistant or fungus-proofed materials shall be used where applicable. DSID's fabricated of materials other than those specified herein and any design changes required due to the use of such materials shall be subject to the approval of the contracting officer.

3.5 Environmental requirements. The DSID shall operate as specified herein without damage (see 6.6.1) when subjected to shock, vibration, temperature, humidity, and altitude extremes, salt fog, water submersion, and fungus as may be encountered in normal field usage when tested under the conditions specified in 3.5.1 and 3.5.2.

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3.5.1 Operating. The DSID shall operate as specified herein without malfunction or damage while exposed to the following conditions:

(a) High temperature (see 4.6.3.5).
(b) Low temperature (see 4.6.3.6).
(c) High/low temperature (see 4.6.3.8).
(d) Altitude (see 4.6.3.9).
(e) Immersion (see 4.6.3.13).

3.5.2 Non-operating. The DSID shall operate as specified herein without malfunction or damage after exposure to the following conditions:

(a) Humidity (see 4.6.3.12).
(b) Temperature shock (see 4.6.3.7).
(c) Shock (see 4.6.3.11).
(d) Vibration (see 4.6.3.10).
(e) Fungus growth (see 4.6.3.17).
(f) Salt fog (see 4.6.3.16).

3.6 Activation sequence. The activation, initiated by the seismic transducer seismometer, shall be amplified and processed by the seismic system. The output of the seismic system shall cause the encoder to turn on the transmitter. The encoder shall then clock out the split phase message format. A pulse at the end of the message shall turn off the transmitter and encoder.

3.7 Seismic system. The seismic system shall consist of a seismometer, threshold signal processor and a seismic bandpass amplifier. The seismic system (without seismometer) shall have a 3 decibel (dB) bandpass of 9 Hertz (Hz) $\pm$ 2 Hz to 45 Hz $\pm$ 6 Hz. The high-end roll-off shall be greater than 9 dB/octave for frequencies greater than 200 Hz.

3.7.1 Seismometer. The seismometer shall be resonant at 14 Hz, and with associated circuitry, detect seismic signals and activate a very high frequency (VHF) radio transmitter.

3.7.2 Signal processor. The signal processor shall include an input circuit which will reject a damped sinewave (bomb burst) type of signal. The signal processor shall provide activation outputs when seismic signals cause its threshold to be exceeded. The threshold level shall be defined as the level required to cause a DSID activation. The enable pulse shall be 1 to 20 milliseconds in length every 13 $\pm$ 3 seconds and

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of sufficient amplitude to actuate the encoder until the seismic signals cease. The seismic processor shall respond to a 14 Hz sinewave input, which exceeds the threshold by 20 dB within the time interval of 1.5 to 3.0 seconds.

3.7.3 Seismic bandpass amplifier. The seismic bandpass amplifier shall have separate fixed gain settings to be called low, medium, and high. The gain shall be such that a sinusoidal 14 Hz seismometer input of 30 millionths of an inch per second peak-to-peak (micro in/sec p-p), 78 micro in/sec p-p, and 190 micro in/sec p-p, respectively, will cause the threshold of the signal processor to be exceeded and result in DSID activation. The sensitivity tolerance shall be within $\pm$ 2 dB at room temperature and within $\pm$ 3 dB over the environmental temperature.

3.8 Encoder-transmitter characteristics. When seismic signals detected by the seismometer satisfy the signal processor criteria for an intrusion detection, DSID shall emit an encoded VHF radio transmission approximately 60 ms long. This message shall include a unit identifier formed by the combination of the preamble, frame mark, message ID, sensor ID, and parity as shown in Figure 1. This combination of parts forms an "identity message" and shall be transmitted in response to an activation. The eighteen-bit message shall be transmitted digitally as split phase frequency-shift keying (FSK) at a bit rate of 300 bits/sec. The bit structure shall be as shown in Figure 2 and is described as follows:

A "1" shall be formed by a 1.667 ms on the mark frequency followed by 1.667 ms on the space frequency: A "0" shall be formed by 1.667 ms on the space frequency followed by 1.667 ms on the mark frequency. The deviation from center frequency shall be plus 3.0 kilohertz (kHz) ($\pm$ 10 percent) for a "mark" and minus 3.0 kHz ($\pm$ 10 percent) for a "space". Rise and fall time between mark and space transition shall not exceed 300 microseconds. The overshoot on the mark or space frequencies shall not exceed the tolerance for those frequencies. Long term bit rate stability (60 days) shall be $\pm$ 3 percent. The short term stability (during a transmission) shall be $\pm$ 0.1 percent. The bit rate tolerance shall be established by the bit rate stability.

3.8.1 Identity message format and bit rate. The Sensor ID code of the identity message format shall be controlled by the setting of the code bar, located on potting surface of the electronics pack by a series of wire loops (see Table I). The code bar, in conjunction with the encoder, shall control the identity message transmitted by the DSID.

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The sensor ID portion of the identity message shall be capable of being set one time only. The code bar identification selections are made by opening one or more of the possible conduction paths for each bit in the message. A "1" is set by continuous wire. A "0" is set by cutting the desired wire on the code bar.

3.8.1.1 Preamble (8 bits). The preamble shall be included with each message type and shall consist of 8 zeros.

3.8.1.2 Frame mark (1 bit). The frame mark shall be included with each message and shall consist of a single "one".

3.8.1.3 Message ID (2 bits). Each message shall contain a message ID code, which shall consist of 2 zeros.

3.8.1.4 Sensor ID and parity (6 + 1 bits). Each message shall contain a sensor ID code as shown in Table I. A parity bit shall be included with the sensor ID code and shall be automatically set by the encoder. This parity bit shall follow the sensor ID and shall be associated with only the sensor ID. The parity bit shall be a "one" if the sum of the one bits in the sensor ID is even and shall be a "zero" if the sum is odd.

TABLE I. ID settings.

ID No.	Bits 12	34	56
1	OS	OS	OS
2	OS	OS	SO
3	OS	OS	SS
4	OS	SO	OS
5	OS	SO	SO
6	OS	SO	SS
7	OS	SS	OS
8	OS	SS	SO
9	OS	SS	SS

ID No.	Bits 12	34	56
11	SO	OS	OS
12	SO	OS	SO
13	SO	OS	SS
14	SO	SO	OS
15	SO	SO	SO
16	SO	SO	SS
17	SO	SS	OS
18	SO	SS	SO
19	SO	SS	SS

"S" INDICATES A SHORTED CONDUCTOR ON NUMBERED CODE BAR.

"O" INDICATES A SEVERED CONDUCTOR ON NUMBERED CODE BAR.

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TABLE I. ID settings.-Continued

ID No.	Bits 12	34	56
21	SS	OS	OS
22	SS	OS	SO
23	SS	OS	SS
24	SS	SO	OS
25	SS	SO	SO
26	SS	SO	SS
27	SS	SS	OS
28	SS	SS	SO
29	SS	SS	SS

ID No.	Bits 12	34	56
31	OO	OS	OS
32	OO	OS	SO
33	OO	OS	SS
34	OO	SO	OS
35	OO	SO	SO
36	OO	SO	SS
37	OO	SS	OS
38	OO	SS	SO
39	OO	SS	SS

ID No.	Bits 12	34	56
41	OS	OO	OS
42	OS	OO	SO
43	OS	OO	SS
44	SO	OO	OS
45	SO	OO	SO
46	SO	OO	SS
47	SS	OO	OS
48	SS	OO	SO
49	SS	OO	SS

ID No.	Bits 12	34	56
51	OS	OS	OO
52	OS	SO	OO
53	OS	SS	OO
54	SO	OS	OO
55	SO	SO	OO
56	SO	SS	OO
57	SS	OS	OO
58	SS	SO	OO
59	SS	SS	OO

ID No.	Bits 12	34	56
61	OO	OO	OS
62	OO	OO	SO
63	OO	OO	SS
64	OO	OS	OO
65	OO	SO	OO
66	OO	SS	OO
67	OS	OO	OO
68	SO	OO	OO
69	SS	OO	OO
71	OO	OO	OO

"S" INDICATES A SHORTED CONDUCTOR ON THE NUMBERED CODE BAR.

"O" INDICATES A SEVERED CONDUCTOR ON THE NUMBERED CODE BAR.

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3.8.2 Transmitter output. The transmitter shall deliver a minimum of 2 watts into a 50-ohms resistive load during the transmission pulse. The transmitter frequency shall be within ± 10 kHz of the assigned center frequency whenever the output power exceeds 200 milliwatts. Within 5 milliseconds after the receipt of the transmission, the transmitter frequency shall be within tolerances of the space frequency, the power output shall exceed the minimum power output requirement, and the message transmission shall have begun. The transmitter power shall fall to less than 200 milliwatts within 5 milliseconds after the trailing edge of the last message bit. Center frequency error shall not exceed ± 30 parts per million (ppm) under any combination of environmental conditions and within the duty factors specified herein.

3.8.3 Channel characteristics. Channels shall be numbered 1 through 1920, beginning at 162.00625 Megahertz (MHz) center frequency, and spaced at 6.25 kHz intervals thereafter. In order to accommodate an inter-channel guard band of 18.75 kHz, only every third channel will be used, beginning with Channel 1 and ending with Channel 1918. This will result in 640 possible VHF channel center frequencies for the DSID. Channel numbers shall be (1 + 3N) where N varies from 0 to 639. Channel number and frequency equivalents shall be as follows:

Channel	Center frequency (In MHz)
1	162.00625
2	162.01250
3	162.01875
4	162.02500
-	—
-	—
-	—
1918	173.98750
1919	173.99375
1920	174.00000

The channels required and the number of DSID units on each channel shall be as specified (see 6.2).

3.9 Antenna. The antenna shall be in accordance with the applicable drawings and as specified herein. The DSID shall employ a 1/4 wave monopole VHF wire type antenna which presents a nominal 50 ohm resistive load, and a maximum Voltage Standing Wave Ratio (VSWR) of 2:1 to the transmitter.

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3.10 Power pack. The power pack shall be in accordance with the applicable drawings and as specified herein. The power pack shall be capable of operating for 60 days with a transmitter duty cycle of 360 activations per day. The power pack shall be nonreplaceable in the field once the DSID has been armed. Power packs shall be interchangeable and usable without modification on all DSID's produced.

3.11 Self-disable. The DSID shall contain a self-disable device capable of permanently disabling the DSID electronics. The DSID shall be controlled by a three-position switch with the functions OFF, TEST, and ARM. No disable shall occur in either the OFF or TEST position. It shall be necessary to sever the ARM looped wire located in the corner of the electronics package on top of the potting surface in order for the ARM position to function. Once placed in the ARM position, a disable shall occur immediately for any of the following conditions:

(a) Recovery. The DSID shall become disabled if the function switch is returned to TEST position or if the power pack is removed.
(b) Voltage sensor. The DSID shall become disabled when the power-pack voltage becomes too low to ensure proper operation.
(c) Life timer. The DSID shall become disabled upon the expiration of the settime and shall be 7 $\pm$ 1 days, 65 $\pm$ 10 days, or end-of-life (EOL). The selection for 7, 65, or EOL shall be at the discretion of the user and shall not be effective until the DSID is in the ARM position.
(d) RF transmission. The DSID shall become disabled if for any reason an RF transmission is longer than the designated identity message width (see 3.8.1). The DSID shall be unable to sustain a spurious oscillation output which may occur in circuitry not controlled by the disabled component.

3.12 External controls. External controls shall be in accordance with the applicable drawing and as specified herein. The single three-position switch positions shall function as follows:

(a) OFF. A state in which no power is applied to any circuit.

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(b) TEST. A state in which power is applied to the detection and transmission circuitry only. No power is applied to the disable circuitry or timer. This position is functional as a sensor but incapable of disabling itself.
(c) ARM. A state in which power is applied as in the TEST position. In addition, all disable circuitry and timer is energized. Operation of the disable circuitry will disable the DSID.

3.12.1 Gain setting. The seismic bandpass amplifier gain setting shall be selected by inserting a designation pin into positions marked Low (L), Medium (M), or High (H) (see 3.7.3).

3.13 Electromagnetic interference. The DSID electromagnetic interference emission and susceptibility characteristics shall conform to the requirements of MIL-STD-461, Class 1, except the radiated spurious emissions shall be at least 40 dB below the level of the fundamental frequency.

3.14 Storage. The DSID, excluding the power pack, shall have a shelf life of not less than 5 years.

3.15 Reliability. The DSID shall have a specified mean-time-between-failure (MTBF) of 15,000 hours and a minimum acceptable MTBF of 3000 hours, assuming 100 activations per day. When specified (see 6.2), reliability testing shall be conducted in accordance with 4.6.3.18.

3.15.1 Failure. A failure is defined as any malfunction that causes or may cause failure to commence operation, cessation of operation, or degradation of performance below levels specified herein.

3.16 Burn-in. Each DSID electronics package shall be operated without failure for 48 consecutive hours. Burn-in shall be performed with the electronics package in the test position.

3.17 Workmanship. Workmanship shall be in accordance with MIL-STD-454, Requirement 9. These requirements applicable to wiring, welding, brazing, plating, riveting, finishes, machine operations, screw assemblies, and freedom of parts from burrs, sharp edges, or any other damage or defect that could make the part or equipment unsuitable for the purpose intended.

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3.17.1 Soldering. All soldering shall be performed in accordance with MIL-STD-454, Requirement 5.

3.17.2 Riveting. Riveted joints shall be tight, joined parts shall be undamaged, and rivet heads shall be properly seated and tight against their bearing surfaces.

3.17.3 Switches and legends. Switches and other movable parts shall operate properly without binding or other undue restrictions. Switches shall be assembled in an alined condition in order to assure accurate registrations with switch positions and fixed markings. Legends shall not deteriorate as a result of normal usage.

3.17.4 Wiring.

3.17.4.1 Protection. Wires and cables shall be placed and protected so as to avoid contact, under specified service conditions, with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insulation material.

3.17.4.2 Splicing. Wires in a continuous run between two terminals shall not be spliced during the wiring operation.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for ensuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.1.2 Disassembly inspection. Failure of any examination or test by the preproduction model shall be cause for disassembly, in the presence of a Government representative, of the preproduction model to the extent necessary to determine the cause of failure.

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Each disassembled part shall be examined in detail for compliance with the specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the preproduction model.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of packaging (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. The DSID shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for performing the inspection specified in 4.1.2.

4.3.2 Tests. The DSID shall be subjected to the tests marked "X" in Column 1 of Table II. Failure of any test shall be cause for performing the inspection specified in 4.1.2.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production DSID's will be selected at random by the Government from the DSID's being produced from production tooling and will be examined for the defects specified in 4.6.1 and subjected to the tests marked "X" in Column 2 of Table II to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production DSID shall not exclude the remaining DSID's from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any and all other tests contained in this specification as part of the initial production test and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

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4.4.1 Inspection failure. Failure of an initial production DSID to meet any requirement specified herein during and as a result of the inspection specified in 4.4 shall be cause for rejection of the initial production DSID and shall be cause for refusal by the Government to continue acceptance of production DSID's until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on DSID's previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all DSID's accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Sampling. Sampling for tests shall be in accordance with MIL-STD-105, Inspection Level II.

4.5.2 Examination. DSID samples selected in accordance with 4.5.1 shall be examined for the possible defects specified in 4.6.1. AQL shall be 1.0 percent defective for major defects and 2.5 percent defective for minor defects.

4.5.3 Tests.

4.5.3.1 Individual. Each DSID shall be subjected to the tests marked "X" in Column 3 of Table II. Failure of any test shall be cause for rejection.

4.5.3.2 Samples. Samples selected in accordance with 4.5.1 shall be subjected to the tests marked "X" in Column 4 of Table II. AQL shall be 1.5 percent defective.

4.6 Inspection procedure.

4.6.1 Examination. The sample DSID's shall be examined as specified in paragraph 4.5.2 for the following possible defects.

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Major

101. Protective treatment not as specified.
102. Materials not as specified.
103. Printed wiring assemblies and boards not as specified.
104. Fungus and moisture resistance not as specified.
105. Requirements for electronics package not as specified.
106. Component parts not as specified.
107. Construction not as specified.
108. Antenna not as specified.
109. Power pack not as specified.
110. Self-disable not as specified.
111. External controls not as specified.
112. Adjustments not as specified.
113. Wiring and cabling not as specified.
114. Case not as specified.
115. Interchangeability not as specified.

Minor

201. Treatment and painting not as specified.
202. Identification and instruction marking not as specified.
203. Workmanship not as specified.

4.6.2 Test schedule. The test schedule is contained in Table II.

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TABLE II. Test schedule.

Preproduction	Initial Production and Inspection Comparison	Quality Conformance Individual	Quality Conformance Samples	Test	Test Paragraph	Requirement Paragraph
1	2	3	4	5	6	7
X	-	X	-	Burn-in	4.6.3.2	3.16
-	-	X	-	Unit test	4.6.3.4	3.6 through 3.9, 3.11, and 3.12
X	X	-	X	High operating temp.	4.6.3.5	3.5.1
X	X	-	X	Low operating temp.	4.6.3.6	3.5.1
X	X	-	X	Temp. shock	4.6.3.7	3.5.2
-	-	-	X	High, low operating temp.	4.6.3.8	3.5.1
X	X	-	-	Altitude	4.6.3.9	3.5.1
X	X	-	-	Vibration	4.6.3.10	3.5.2
X	X	-	-	Shock	4.6.3.11	3.5.2
X	X	-	X	Humidity	4.6.3.12	3.5.2
X	X	X	X	*Immersion	4.6.3.13	3.5.1
X	X	X	-	External controls	4.6.3.14	3.12
X	X	-	-	Electromagnetic interference	4.6.3.15	3.13
X	X	-	-	Salt fog	4.6.3.16	3.5.2
X	-	-	-	Fungus growth (Test A)	4.6.3.17.1	3.5.2 (e)
-	X	-	-	Fungus growth (Test B)	4.6.3.17.2	3.5.2 (e)
X	X	-	-	Reliability	4.6.3.18	3.15

*Immersion test shall be performed after all the environmental tests, except fungus, have been completed.

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4.6.3 Tests.

4.6.3.1 Conditions. Unless otherwise specified herein, all tests shall be conducted at an ambient room temperature of 75° F ± 5° F with the power "on". The Government will loan one or more Portatales AN/USQ-46 for test purposes. The exact number of sets to be loaned will be as specified (see 6.2 and 6.5). For "power on" tests, an external power source shall be used.

4.6.3.2 Burn-in. Each DSID shall be subjected to a 48-hour burn-in. During burn-in, the DSID shall be exercised through all of its functions 100 times per each 24-hour period. Failures during burn-in shall be analyzed, and corrective action taken when appropriate. Copies of failure reports shall be forwarded to the contracting officer. Only DSID units that successfully complete the 48-hour burn-in shall be subjected to the testing specified in 4.6.2.3.

4.6.3.3 Test equipment calibration. Calibration of test equipment shall be in accordance with MIL-C-45662.

4.6.3.4 Unit test. Preliminary to this test, the geophone resistance shall be checked. The resistance shall be 7800 ohms ± 20 percent (6240 9360 ohms). Power supplies capable of adjustment from +5.0 to +7.0 volts and from 18.0 to 30.0 volts shall be provided for testing DSID characteristics. Tests shall be conducted at both 23.0 volts and 28.0 volts on the high voltage supply and at both 5.5 volts and 6.8 volts on the low voltage supply. Standby currents shall be ≤ 200uA for the 28-volt supply and ≤ 110uA for the 6.8-volt supply. Transmit currents shall be not less than 375 ma and not more than 675 ma.

(a) With the power sources connected and using the AN/USQ-46 Portatale, seismically stimulate the geophone and observe the DSID for proper activation sequence (see 3.6).
(b) Substituting a signal generator for the geophone, test the seismic amplifier for gain in the pass band and for rolloff at 200 Hz. Tests shall be made to determine the ability of the DSID to detect seismic signals, and to activate the radio transmitter (see 3.7).

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(c) A test shall be conducted to assure that the transmitter output power is $\geq$ 2 watts for the duration of the 60-millisecond output and that the transmitter frequency is within $\pm$ 10 kHz of the assigned frequency when the power supply voltage is at the minimum and maximum test voltages. The message bit rate, format, length of transmission, and signal characteristics shall be checked for conformance to 3.8.

(d) The self-disable system shall be tested for proper operation under conditions of recovery and low power supply voltage. With the DSID connected to the adjustable supply voltage and an oscilloscope monitoring the disable pulse test point, a disable pulse shall be observed when:

(1) The 6.8-volt power supply is reduced to 5.4 $\pm$ 0.35V.
(2) The 28.0-volt power supply is reduced to 19 $\pm$ 1.0V.
(3) The function switch is returned to the TEST position from the ARM position.
(4) The end-of-life voltage is greater than 10 mv (see 3.11) when measured between the EOL terminal and ground.

(e) The control switch shall be checked for operation in each of its 3 positions and shall function with appropriate stops and without binding (see 3.12).

Nonconformance to 3.6, 3.7, 3.8, 3.9, 3.11, or 3.12 shall constitute failure of this test.

4.6.3.5 High temperature (power on). The DSID shall be subjected to the high temperature test in accordance with MIL-STD-810, Method 501.1, Procedure I. The highest operating temperature shall be 145° F. Any evidence of damage (see 6.6.1) or inability to perform as specified in 4.6.3.4 shall constitute failure of this test.

4.6.3.6 Low temperature (power on). The DSID shall be subjected to the low temperature test in accordance with MIL-STD-810, Method 502.1, Procedure I. The lowest temperature shall be -54° C (-65° F) maintained

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for 8 hours. The operating temperature shall be 32° F for 8 hours. Any evidence of damage (see 6.6.1) or inability to perform as specified in 4.6.3.4 shall constitute failure of this test.

4.6.3.7 Temperature shock (power off). The DSID shall be subjected to the temperature shock test in accordance with MIL-STD-810, Method 503.1, Procedure I. Any evidence of damage (see 6.6.1) or inability to perform as specified in 4.6.3.4 shall constitute failure of this test.

4.6.3.8 High/low operating temperature. The DSID, with power on only at operating temperature ranges, shall be subjected to the following high and low temperature cycle shock tests. Any evidence of damage (see 6.6.1) or inability to perform as specified in 4.6.3.4 shall constitute failure of this test.

(a) Place DSID in test chamber and activate. Raise the temperature to +145° F and maintain for four (4) hours. Perform an abbreviated operational test and deactivate the DSID.
(b) Raise the temperature to +160° F and maintain for four (4) hours.
(c) Reduce the temperature at a rate of not less than 45° F per minute to -65° F and maintain for four (4) hours.
(d) Raise the temperature at a rate of not less than 45° F per minute to +160° F and maintain for four (4) hours.
(e) Reduce the temperature to +32° F and maintain for one (1) hour. At this point activate the DSID and test maintaining the temperature at +32° F for three (3) hours. Perform an abbreviated operational test.
(f) Raise the temperature to ambient room temperature and maintain for one (1) hour. At the conclusion of this test, the DSID shall be subjected to the tests specified in 4.6.3.4.

4.6.3.9 Altitude (power on). The DSID shall be subjected to the altitude test in accordance with MIL-STD-810, Method 500.1, Procedure I, except that the operating altitude shall be 14,000 feet. Any evidence of damage (see 6.6.1) or inability to perform as specified in 4.6.3.4 shall constitute failure of this test.

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4.6.3.10 Vibration (power off). The DSID shall be removed from its shipping container and subjected to the vibration test in accordance with MIL-STD-810, Method 514.2, Procedure X, Curve AX, Figure 514.2-7. Any evidence of damage (see 6.6.1) or inability to perform as specified in 4.6.3.4 shall constitute failure of this test.

4.6.3.11 Shock (power off). The DSID shall be subjected to the following shock tests in accordance with MIL-STD-810.

4.6.3.11.1 Method 516.2, Procedure I, Figure 516.2-1, Amplitude b, Time Duration d (power off). The DSID shall be removed from its shipping container during this test. Any evidence of damage (see 6.6.1) or inability to perform as specified in 4.6.3.4 shall constitute failure of this test.

4.6.3.11.2 Method 516.2, Procedure V (power off). The DSID shall be removed from its shipping container during this test. Any evidence of damage (see 6.6.1) or inability to perform as specified in 4.6.3.4 shall constitute failure of this test.

4.6.3.12 Humidity (power off). With case open, the DSID shall be subjected to the humidity test in accordance with MIL-STD-810, Method 507.1, Procedure I with the following exception: Sample DSID (see 4.5.3.2) shall be subjected to only 5 cycles of Steps 2, 3, and 4 of Procedure I. Any evidence of damage (see 6.6.1) or inability to perform as specified in 4.6.3.4 shall constitute failure of this test.

4.6.3.13 Immersion (power on). The DSID shall be subjected to the leakage test in accordance with MIL-STD-810, Method 512.1, Procedure I, except that the temperature difference between the water and the DSID at the time of submersion shall be not less than 10° C. The water shall be cooler. The antenna base constitutes the uppermost point of the unit. Any evidence of damage (see 6.6.1), water leakage, or inability to perform as specified in 4.6.3.4 shall constitute failure of this test.

4.6.3.14 External controls. The DSID switch shall be subjected to a test of all its functions. Any evidence of mechanical or electrical damage or inability to perform as specified in 4.6.3.4 shall constitute failure of this test.

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4.6.3.15 Electromagnetic interference (power on). The prototype models equipped for the reduction of electromagnetic interference in accordance with 3.13 shall be tested in accordance with MIL-STD-461, except that the radiated harmonic components shall be 40 dB below the fundamental frequency. The contractor shall furnish the contracting officer the test report required by MIL-STD-461 within the time interval specified (see 6.2). Upon approval of the report by the contracting officer, and provided all other rquirements of the specification are met, these models shall be used as standards for all production DSID's. Disapproval of any report shall constitute failure of this test.

4.6.3.16 Salt fog (power off). The DSID with all covers, seals, and protective devices removed as required for maintenance settings or adjustments, shall be subjected to the salt fog test in accordance with MIL-STD-810, Method 509.1, Procedure I. Any evidence of damage or inability to perform as specified in 4.6.3.4 shall constitute failure of this test.

4.6.3.17 Fungus growth.

4.6.3.17.1 Fungus growth (Test A). The DSID shall be placed within the Tropical Testing Chamber, USAMERADCOM, Fort Belvoir, Virginia (see 6.4) and subjected to exposure under simulated cyclic tropical conditions prevailing therein for a period of 90 days. Environmental conditions within the chamber are as follows:

> The Tropical Testing Chamber is operated on a daily temperature humidity cycle typical of a tropical area. For an 8-hour period, the temperature is held at 85° F and the relative humidity at 70 percent. During the next 8 hours of the cycle, the temperature is gradually lowered to 75° F, and the relative humidity are then kept at 75° F and 95 percent for a 4-hour period, followed by a gradual change to 85° F and 70 percent during the next 4-hour period, which completes the cycle.

The chamber atmosphere is heavily charged with spores of various species of fungi and provides continuous inoculation of exposed items and materials to such an extent that artificial prior inoculation with fungus spores is unnecessary.

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The following list is representative of the fungi present in the chamber:

Acremonium alternatum	Memoniella echinata
Ascotricha xylina	Mucor sp.
Aspergillus fluvus	Myrothecium verrucaria
Aspergillus niger	Nygrospora sphaerica
Aspergillus oryzae	Penicillium citrinum
Aspergillus versicolor	Penicillium luteum
Aspergillus terreus	Penicillium purpurogenum
Chaetomium bostrychodes	Penicillium stoloniferum
Chaetomium indicum	Pollularia sp.
Chaetomium globosum	Rhizopus nigricans
Coniothyrium sp.	Spicaria sp.
Curvularia sp.	Stachbotrys atra
Fusarium sp.	Trichoderma viride

Upon removal from the chamber after the 90-day exposure period, the DSID shall be examined for damage and tested in accordance with 4.6.3.4. Any evidence of damage or inability to perform as specified in 4.6.3.4 shall constitute failure of this test.

4.6.3.17.2 Fungus growth (Test B). The DSID shall be tested in accordance with MIL-STD-810, Method 508.1, Procedure I. After conclusion of the test, the DSID shall be examined for damage (see 6.6.1). The DSID shall then be subjected to all tests specified in Column 3 of Table II. Inability to pass any of these tests or evidence of deterioration (see 6.6.1) shall constitute failure of the fungus test.

4.6.3.18 Reliability test. When specified (see 3.15), the DSID shall be tested as specified in 3.15 in accordance with MIL-STD-781, Test Plan VI. When this reliability test is to be conducted, the time frame for conducting the test, the number of DSID units to be tested, and who shall perform the tests shall be as specified (see 6.2). The DSID shall be activated 100 times per day during the reliability test. Accept/reject criteria shall be as specified in MIL-STD-781, Test Plan VI. Items selected for reliability testing shall have been "burned-in" as specified in 3.16 and shall have passed the quality conformance inspection specified in 4.5. Periodic performance checks shall be made throughout the reliability test. The reliability test shall be conducted under ambient environmental conditions. Nonconformance to 3.15 shall constitute failure of this test.

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4.7 Inspection comparison. The Government may select DSID's at any time during the contract production period and subject these DSID's to the examination specified in 4.6.1 and the tests marked "X" in Column 2 of Table II to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production DSID. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison DSID to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison DSID and shall be cause for refusal by the Government to continue acceptance of production DSID's until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on DSID's previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all DSID's accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging.

4.8.1 Quality conformance inspection of pack.

4.8.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.8.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.1.3 Examination. Samples selected in accordance with 4.8.1.2 shall be examined for the following possible defects. AQL shall be 1.0 percent defective for major defects and 2.5 percent defective for minor defects.

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Major

116. Materials, methods, and containers not as specified for Level A or B. Each incorrect material, method, or container shall be considered one defect.
117. Items not cushioned and secured within the box as specified for Level A.
118. Fiberboard box or polystyrene container not sealed or secured as specified for Level A.
119. Quantities packed together exceed the size or weight limitation of the box for Level A or B.
120. Strapping not as specified for Level A.
121. Items of unlike description packed together for Level A, Level B, or Commercial.

Minor

204. Marking missing, illegible, incorrect, or incomplete for Level A, Level B, or Commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Each electronics package and each power pack shall be individually placed in a box conforming to PPP-B-636, W6c. The item shall be cushioned and secured within the box in a manner to prevent free movement or damage by the use of polystyrene or fiberboard inserts. Fiberboard inserts shall be fabricated from the same material as the box. Polystyrene inserts shall conform to MIL-P-60312, Type II, Class 1, or PPP-C-850, Type I, Class 3. The box shall be closed and sealed as specified for Method V in the appendix to the box specification. As an alternate, each electronics package and each power pack may be individually placed in a two-piece molded polystyrene container. The interior configuration of the container shall be designed to prevent movement of the item within the container. The container shall be secured to prevent opening by the use of tape conforming to PPP-T-97, Type I.

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5.1.2 Commercial. Each electronics package and each power pack shall be individually preserved in accordance with FED. STD. No. 356.

5.2 Packing. Packing shall be Level A, Level B, or Commercial, as specified (see 6.2).

5.2.1 Level A. The electronics packages and the power packs in equal number, preserved as specified in 5.1, shall be packed in close-fitting boxes conforming to PPP-B-601, Overseas Type, Grade B, style optional. Only items of like descriptions shall be packed together. The quantities per box shall not exceed the weight limitation of the box. Box closure and strapping shall be in accordance with the appendix to the box specification. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be Finish B. When specified (see 6.2), strapping shall be Finish A.

5.2.2 Level B. The electronics packages and power packs, preserved as specified in 5.1, shall be packed as specified in 5.2.1, except boxes shall be Domestic Type. As an alternate, boxes conforming to PPP-B-636, V3c, V11c, V13c, or V15c may be used when the quantity to be packed does not exceed the gross weight or size limitation of the box. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.2.3 Commercial. The electronics packages and the power packs, preserved as specified in 5.1, shall be packed in accordance with FED. STD. No. 356. Only items of like description shall be packed together.

5.3 Marking.

5.3.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial packaging. Marking shall be in accordance with FED. STD. No. 356.

6. NOTES

6.1 Intended use. The DSID is intended for detection of seismic signals resulting from movement of vehicle and personnel on foot. The DSID is a self-contained, hand-emplaced sensor designed to detect the

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presence of seismic disturbances as a result of personnel or vehicle movement. The DSID performs a signal signature analysis in order to isolate those disturbances that have specific characteristics. When a disturbance is detected, an RF transmitter is activated and emits an identification (ID) code which identifies the specific DSID, in addition to announcing the detection of a seismic disturbance. The DSID is protected against enemy use and cannot be recovered.

6.2 Ordering data. Procurement documents shall specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of preproduction model and number of DSID's required (see 3.2).
(c) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some, but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
(d) When initial production inspection is required, and number of DSID's to be furnished (see 3.3).
(e) Channels required and the number of DSID units to be furnished on each channel (see 3.8.3).
(f) When reliability test is required, the time frame for conducting the reliability test, the number of DSID units to be tested, and who shall perform the test (see 3.15 and 4.6.3.18).
(g) The number of Frequency Monitor Sets that the Government will loan for test purposes (see 4.6.3.1 and 6.5).
(h) Time interval for submission of electromagnetic interference test plans and test reports (see 4.6.3.15).
(i) Degree of preservation and packing required (see 5.1 and 5.2).
(j) When strapping shall be other than as specified (see 5.2.1).

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6.3 Preproduction model. Any changes or deviations of production DSID units from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish DSID units conforming to this specification.

6.4 Tropical Testing Chamber. Arrangements should be made to make the Tropical Testing Chamber, USAMERADCOM, Fort Belvoir, Virginia available to the contractor for conducting the fungus growth test (see 4.6.3.17).

6.5 Government-loaned property. The contracting officer should arrange to loan the property specified in 4.6.3.1.

6.6 Definitions.

6.6.1 Damage. Damage is deformation, corrosion, loosening of parts, breakage, or change of fit of any component or part, or any physical condition resulting in noncompliance of the DSID with the provisions of this specification.

Custodians:
Army - ME
Navy - SH

Preparing activity:
Army - ME

Project 6350-0037

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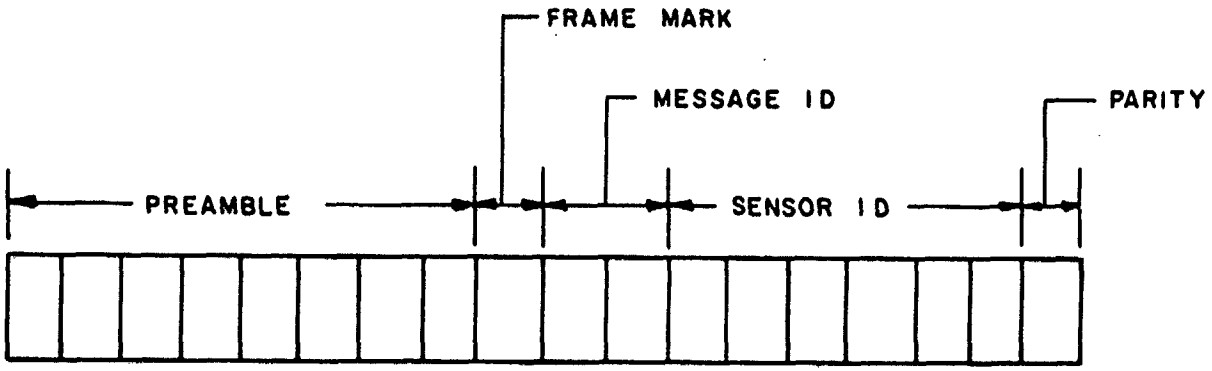


Figure 1. Identity message format.

X-2767

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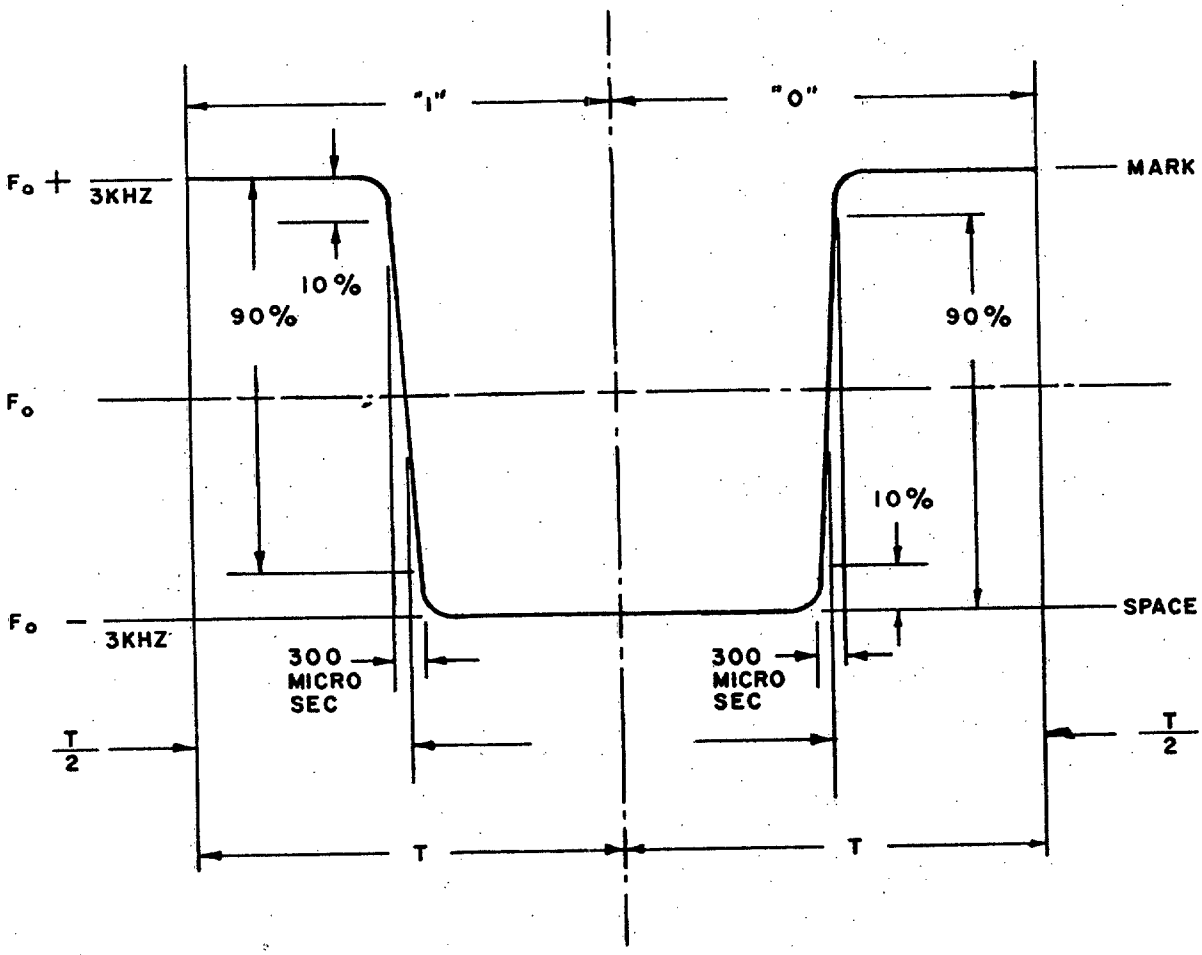


Figure 2. Message bit structure.

X-2768

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DOCUMENT IDENTIFIER AND TITLE: MIL-A-52832A Alarm Set, Anti-Intrusion, Restricted Area: AN/GSQ-159 (DSID)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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